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METHODOLOGY · VERSION 5.0

Guilford County Neighborhood Vitality Index

A Block-Group Market Segmentation with Outcome-Criterion Variable Selection

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ABSTRACT

This proposal outlines a methodology for the Guilford County Neighborhood Vitality Index (GCNVI) as a block-group market segmentation built through outcome-criterion variable selection. Where the original composite combined indicators under theory-derived domain weights, this revision proceeds in the methodological tradition of Mallach (2008, 2012) and Sills (2015), in which neighborhoods are scored against an empirical structure of outcomes the index is designed to predict.

Four criterion variables anchor the model: three measures of intergenerational mobility from the Opportunity Atlas (Chetty et al., 2018, 2026) and the composite Child Opportunity Index 3.0 (Noelke et al., 2024). Approximately one hundred candidate predictors drawn from federal, state, county, parcel, and partner-collected data are standardized as z-scores, screened by Pearson correlation against the four criteria, and submitted to regression models with variance inflation factor diagnostics for the refinement of a parsimonious set of structurally significant indicators. The final summative index of standardized scores will be assessed for internal consistency via Cronbach's alpha, partitioned into five categories by visual binning (Mean $\pm$ SD) in SPSS, and mapped from red to green across all 338 block groups in Guilford County (2020-vintage frame).

The procedure is unweighted, transparent, reproducible, and directly comparable across the county. We argue that this design represents a defensible methodological pivot from the theory-weighted composite, although it carries its own constraints, most importantly the embedded normative choice of criterion outcomes and the well-documented small-area uncertainty of American Community Survey estimates at block-group geography (Spielman et al., 2014). These constraints are addressed through margin-of-error filtering, sensitivity analysis under alternative criterion specifications, and triangulation against three external composites.

VERSION HISTORY

v2.0 (June 9, 2026): methodology proposal — outcome-criterion pivot. v3 (June 23, 2026): full variable catalog and analytic procedure. v5 (July 7, 2026, this edition): documents the completed v5 data pipeline — the full criterion battery is assembled (Y1 337/338, Y2 337/338, Y3 337/338, Y4 338/338 block groups; 21 of 21 automated verification checks passed) on the 338-block-group 2020 frame — and specifies the v6 expansion of Domain I (dual-source crime measurement) and Domain J (civic infrastructure and social capital), the addition of eviction filings (E10) and the Opportunity Atlas incarceration measure (I7), and the associated limitations (Section 10.7).

1. Introduction: From a Theory-Weighted Composite to a Data-Driven Segmentation

When we began to investigate possible models for a Guilford County Neighborhood Vitality Index, we started with frameworks that utilized theory-derived domain weights. The present revision proceeds from a different methodological commitment. It is responsive to two considerations: the need for a Census block group level analysis and the empirical record on theory-derived weights themselves. Although theory-derived weights are transparent in a way that factor-analytic weights are not (see previous IRI document “Measuring Neighborhood Disadvantage & Opportunity: A Literature Review of Indices for Targeting Economic Development in Historically Disinvested Communities”), they remain contestable, and the argument over which domain deserves more weight than another is exactly the argument the Coalition of Anchors does not need to have repeatedly.

Selecting variables instead by their empirical correlation with a set of pre-specified outcome criteria shifts the disputed question from “how much should each domain matter” to “which structural conditions of block groups predict the outcomes one cares about,” which is a question the data can answer. Therefore, the present version of the Guilford County Neighborhood Vitality Index is based on the market segmentation tradition of Alan Mallach (2008, 2012), which has shaped neighborhood-stabilization work in Newark, Youngstown, Detroit, and even the High Point Core City (Sills 2015). In urban planning and community development, the market segmentation model is a data-driven framework designed to categorize city neighborhoods into distinct typologies. This allows local governments to tailor highly specific interventions to the unique conditions of different geographic areas.

The four criterion outcomes have been specified in advance: 1) the median household income at age 35 of children born to parents at the 25th percentile of the national income distribution who grew up within a block group; 2) the share of such children who reach the top quintile of the adult income distribution across all parental income levels; 4) the employment rate at age 35 of children who grew up in the block group; and, 4) the composite Child Opportunity Index 3.0 score. The first three are direct measures of intergenerational economic mobility from the Opportunity Atlas (Chetty et al., 2018, 2026); the fourth is a cross-sectional measure of the present-day opportunity structure facing children growing up in the block group today (Noelke et al., 2024). We treat all four as criterion variables on the substantive judgment that the purpose of the Atlas is to make legible the structural conditions of neighborhoods that condition the life chances of the children who live in them; a commitment that follows directly from Wilson’s (1987) framing of concentrated disadvantage and from Sharkey’s (2013) extension of that argument across generations.

The procedure proposed has six steps. We will summarize them here briefly and elaborate them at length in Section 7. First, every candidate predictor is standardized to a z-score against the Guilford County block-group distribution. Second, Pearson product-moment correlation coefficients are

computed between each candidate predictor and each of the four criterion variables. Third, predictors that fail to reach statistical significance against any of the four criteria are excluded from further consideration. Fourth, the surviving predictors are submitted to multivariate ordinary least squares regression models predicting each criterion in turn, with variance inflation factors computed to detect multicollinearity; predictors with VIF values above conventional thresholds (Tabachnick & Fidell, 2019) are excluded, and predictors whose standardized partial regression coefficients are not statistically significant across the four models are also excluded. Fifth, the final retained set of z-scored predictors is summed without further weighting to produce the GCNVI score for each block group. Sixth, Cronbach's alpha is computed on the retained set as a measure of internal consistency (Cronbach, 1951; Nunnally & Bernstein, 1994; Sijtsma, 2009). The resulting block-group scores are then partitioned into five categories by visual binning at $\text{Mean} \pm 0.5 \text{ SD}$ and $\text{Mean} \pm 1.5 \text{ SD}$ in SPSS, mapped from red to green across all 338 block groups in Guilford County (2020-vintage frame), and accompanied by the three external composite comparators.

We want to be clear at the outset about what this does and does not represent. It is not a rejection of the conceptual scaffolding originally proposed by Girard Newkirk. The domains examined herein will include economic production and circulation, household economic position, human capital and opportunity, built environment and housing, health and civic conditions. These domains are the theoretical structure of what neighborhood vitality means, and the variable catalog of this document covers all of them. What changes is the principle by which a composite score is constructed: not the theoretical weighting of indicators but the actual empirical relationship between these indicators and outcomes instead. This presents a more rigorous course, although we will argue later that it carries costs as well as benefits and that the two approaches may be more complementary rather than alternative.

The remainder of this document is organized as follows. Section 2 situates the proposal within the market segmentation tradition in neighborhood research, drawing on the lineage from Mallach's Newark and Youngstown work through the geodemographics literature and into the housing-market typology studies Dr. Stephen Sills and his colleagues at the UNCG Center for Housing and Community Studies conducted in High Point in 2015 (and later in Rocky Mount, Greensboro, and Baton Rouge). Section 3 articulates the logic of criterion-validated variable selection and its position within the broader literature on composite index construction. Section 4 examines the four criterion variables in detail and defends their selection. Section 5 introduces the candidate predictors and the conceptual domains from which they are drawn. Section 6 presents the variable catalog itself with approximately one hundred candidate measures organized into thirteen substantive domains, each annotated with operationalization, data source, geographic resolution, and rationale. Section 7 sets out the analytic procedure in detail. Section 8 compares this method against six alternatives in the composite-index literature and acknowledges what each method captures that the others do not. Section 9 lays out the implementation pathway, including the parcel-to-block-group aggregation rules, the margin-of-error filtering protocol, and the

integration with the Newkirklytics field-collected indicators. Section 10 acknowledges the limitations of the proposal.

2. The Market Segmentation Tradition in Neighborhood Research

The methodological tradition into which this proposal places the GCNVI is the housing-market segmentation approach that Alan Mallach created to systematize data-informed, neighborhood-level interventions over the past two decades. In Mallach's (2008, 2012, 2018) treatment, market segmentation is the practice of partitioning a city's residential geography into a small number of types distinguished by a parsimonious set of housing-market and socioeconomic conditions, and tying each type to a corresponding set of policy interventions appropriate to its conditions. The tradition has roots in the planning practice of geodemographics (Burton, 2002; Harris et al., 2005) but acquired its contemporary form in Mallach's work in Newark in the early 2000s and in the Youngstown 2010 plan that followed (Mallach, 2008). Its central conceptual claim is that the policy environment differs categorically across neighborhood types: a neighborhood whose housing stock is structurally sound but whose vacancy rate is rising calls for stabilization interventions of a kind that would be inappropriate in a neighborhood whose stock is deteriorating and whose vacancy rate is high, and that would be doubly inappropriate in a neighborhood whose stock is sound and whose vacancy rate is low. Categorical typologies, the Mallach argument runs, communicate this differentiated policy logic in a way that continuous rankings cannot.

The 2015 Market Segmentation Report prepared for the City of High Point Community Development and Housing Department (Sills, 2015) applied Mallach's typology framework to thirty-two block groups in High Point's Core City. The methodology used Pearson correlations to select variables from a larger candidate set; converted retained variables to z-scores to standardize across scales; constructed a summative composite; partitioned the composite into five categories (Stable, Functioning, Constrained, Weak, and Extremely Weak) by reference to the distribution of scores across block groups. Tied to each category was a defined set of intervention recommendations consistent with Mallach's policy logic (and that of the Center for Community Progress). The 2015 study did not employ outcome-criterion variable selection in the formal sense this proposal develops, but it shared the core methodological commitments that distinguish the market segmentation tradition from the deprivation-index tradition: block group rather than tract as the unit of analysis, categorical rather than continuous output as the final form, and the assumption that the index exists to support differentiated parcel- and block-level intervention rather than to rank-order need for redistribution.

The market segmentation tradition has been criticized on several grounds, and any responsible restatement of the GCNVI in its terms must engage those criticisms. The first criticism is that block-group geographies are statistically unstable for many of the indicators that compose neighborhood

typologies. Spielman et al. (2014) documented systematically that American Community Survey estimates at the block-group level frequently carry margins of error large enough that ranking block groups on those estimates produces statistically indefensible orderings. The Hannan et al. (2023) reproducibility critique of the Area Deprivation Index (ADI) raises a related concern: a composite constructed from unstable inputs inherits their instability, and small differences in scoring procedure or vintage can produce substantively different rankings. In Section 7 we address these concerns with the proposal of using margin-of-error filtering and sensitivity-analysis protocols.

The second criticism of the market segmentation tradition concerns the embedded normative choice of category cutpoints. Visual binning at $\text{Mean} \pm 0.5 \text{ SD}$ and $\text{Mean} \pm 1.5 \text{ SD}$ produces a roughly quintile-shaped distribution under the assumption of approximate normality, but for skewed score distributions the cutpoints can produce categories of substantively misleading sizes. The 2015 High Point study's category labels (Stable through Extremely Weak) also carry normative weight that the underlying continuous score does not, and analysts working in the Mallach tradition have long debated whether those labels guide policy in productive or stigmatizing directions (Mallach, 2018; Hackworth, 2016). We propose in Section 7 to retain visual binning for the mapped output while preserving the underlying continuous z-score sum for analytical purposes, so that the categorical mapping serves communication while the continuous score serves the regression-based refinement and the alpha-based reliability assessment.

The third criticism concerns the relation between the market segmentation approach and the deprivation-index tradition with which it shares conceptual territory. The Townsend (1988) Material Deprivation Index, the Singh (2003) Area Deprivation Index, and the Distressed Communities Index (EIG, 2025) all operate at the tract level, weight their inputs by formal procedure, and produce continuous scores intended for nationwide comparison. They do not partition geography into intervention typologies, and they do not normally proceed by outcome-criterion variable selection. Sills has argued elsewhere (Sills, 2025) that the market segmentation and deprivation-index traditions are complements rather than competitors as they answer different questions and serve different decision contexts. The deprivation indices answer questions about relative need across a national distribution while the market segmentation indices answer questions about differentiated intervention within a particular jurisdiction. The Coalition of Anchors and the Guilford County partners face the second kind of question, and the methodological restatement of the GCNVI follows from that fact.

There is a fourth body of literature on which we draw, more recently developed than the Mallach lineage but adjacent to it: the outcome-criterion variable selection approach that Knighton et al. (2016) employed in the Stanford Reproducible Area Deprivation Index (ADI) and that the Stanford team has subsequently extended in their work on small-area socioeconomic position. The Stanford approach uses health and economic outcomes as criterion variables against which candidate predictors are validated, with the result that the retained variable set carries explicit predictive validity for the outcomes the index is intended to support. The procedure follows the Stanford logic closely, with the

substitution of the four child-mobility and child-opportunity criteria for the Stanford team's mortality and morbidity outcomes. The substitution is substantive as it reflects the substantive judgment that the Atlas is built to serve the long-term economic-mobility outcomes that Wilson (1987), Sampson (2012), Sharkey (2013), and Chetty and Hendren (2018) have identified as the central neighborhood-effect outcomes for which spatial inequality matters.

2.1 Why treat this as a methodological pivot rather than a rejection

The five-domain conceptual structure of earlier drafts included economic production and circulation, household economic position, human capital and opportunity, built environment and housing, health and civic conditions. This remains the architecture of Sills's substantive thinking about what vitality means, and every variable in the catalog below can be assigned to one of those five domains. The Newkirknomics commitments (Newkirk, 2025; Newkirklytics, 2025) that placed the largest weight on economic production and circulation also remain conceptually intact and the variable catalog will include the full set of Neighborhood Economy Index (NEI) and Newkirknomics Benchmark Framework (NBF) indicators that Girard Newkirk (Newkirklytics) is operationalizing through the pilot, and we expect that several of them will survive the correlational and regression-based selection procedures.

What changes is the procedural rule that determines which variables ultimately compose the index and how their scores combine. The theory-derived domain weights will be replaced by the empirical structure of correlations and partial regression coefficients against the four criterion outcomes. The argument structure shifts from "these conditions, weighted thus, define vitality" to "these conditions, present in this neighborhood, predict the long-run economic outcomes of the children who grow up here." We take the second claim to be the more defensible one to commit the Coalition of Anchors and Guilford County partners to, and the more useful one for the specific decisions the Atlas exists to support.

3. The Logic of Criterion-Validated Variable Selection

The general theory of criterion-related validity dates to Cronbach and Meehl's (1955) foundational paper on construct validation, in which they distinguished criterion-related validity (the demonstration that a measure correlates with a specified external outcome) from content validity, predictive validity, and the broader construct validity that comprehends both. In the contemporary composite-index literature, criterion-related variable selection has been operationalized most fully in the Stanford Reproducible ADI (Knighton et al., 2016), in which the variable set was retained on the basis of its demonstrated correlation with downstream health outcomes, and in adjacent applications of the same logic to neighborhood-effect research more broadly (Diez Roux & Mair, 2010; Galster, 2012). The procedure has three properties we want to make explicit, because they distinguish it from both the theory-derived weighting tradition and the factor-analytic tradition.

The first property is that the criterion variables themselves embed a normative judgment about what the index is for. To choose intergenerational economic mobility and present-day child opportunity as the four criteria is to commit the index to the substantive position that what makes a neighborhood vital is its capacity to generate or sustain economic and developmental opportunity for the children who grow up in it. This is not a neutral choice. It departs from the deprivation-index tradition, which treats material poverty as the construct of interest in itself; from the social vulnerability tradition (Flanagan et al., 2011), which treats disaster preparedness as the construct of interest; and from the Newkirknomics tradition (Newkirk, 2025), which would treat economic production capacity as the construct of interest.¹ Each of these substantive positions could be operationalized through its own set of criterion variables, and the resulting indices would differ. We make the choice of child-mobility and child-opportunity criteria explicit so that the substantive commitment behind the variable selection is legible rather than hidden.

The second property is that criterion-validated selection makes the predictive validity of the index a feature of its construction rather than a question to be tested after the fact. Because variables that fail to correlate significantly with the four criteria are excluded from the composite, the surviving variable set is, by construction, the set of structural conditions that empirically predict the outcomes the index exists to support. This is methodologically attractive but creates a known interpretive constraint that we will return to in the limitations section: the resulting index cannot be used to test the predictive relationship between neighborhood structure and child outcomes, because that relationship has been embedded in its construction. The GCNVI will be a tool for allocation and intervention design, not a research instrument for studying the causal effects of neighborhood conditions on outcomes.

The third property is that the procedure substitutes one set of methodological choices for another, rather than eliminating choice. The variables we include as candidates for selection are still choices; the operationalization of each variable is still a choice; the significance threshold for correlation-based selection ($p < 0.05$ in the primary analysis with a Bonferroni-adjusted threshold reported as sensitivity), the variance inflation factor threshold ($VIF > 5$ as the primary threshold with $VIF > 10$ reported as sensitivity), and the binning rule ($\text{Mean} \pm 0.5 \text{ SD}$ and $\pm 1.5 \text{ SD}$) are all choices we specify in advance. The composite is unweighted only in the narrow sense that the retained variables are summed without further differentiation. The full procedure is filled with consequential decisions, and the rigor of the index depends on making each of those decisions transparently and recording them so that other analysts could reproduce the result.

¹ It is further argued that one must better define a neighborhood not as a place of economic production capacity, but as the foundational building block of society, uniting geographic proximity with social connection to create a physical space for daily life while meeting essential human needs for security, community, and shared resources. It clusters daily necessities like parks, schools, and shops within reach while fostering belonging and improving quality of life.

3.1 What criterion-validated selection is not

Criterion-validated selection is not the same as data-driven dimensionality reduction. Principal components analysis, exploratory factor analysis, and confirmatory factor analysis all operate on the internal correlational structure of the candidate predictor set, identifying latent constructs that account for shared variance among the observable variables. The variable weights derived from these procedures are functions of the correlational structure of the predictors themselves, not of any external outcome. The Singh (2003) ADI weights were derived in this way, and Petterson's (2023) critique of the contemporary ADI rests on the observation that the weights remained fixed from Singh's 1990 census derivation despite the predictor distributions having shifted substantially in the intervening decades. Criterion-validated selection escapes this kind of staleness because the validity of the variable set is tied not to its own internal correlations but to its correlation with external outcomes, which are themselves available for refresh.

Criterion-validated selection is also not the same as expert-elicited weighting. The Robert Wood Johnson County Health Rankings model (UWPHI, 2014) weights health behaviors at 30%, clinical care at 20%, social and economic factors at 40%, and physical environment at 10% on the basis of expert judgment about the relative contribution of each domain to length and quality of life. This is a defensible procedure. But it commits the index to a particular reading of expert consensus, and it produces a composite whose weights cannot be easily revised in response to local empirical conditions. Criterion-validated selection produces weights (or rather, an unweighted summative composite of empirically selected predictors) that derive from the local data themselves and that update naturally when the data are refreshed.

4. The Four Criterion Variables

We propose four criterion variables against which all candidate predictors are screened. Three are drawn from the Opportunity Atlas (Chetty et al., 2018, 2026), and one is the composite Child Opportunity Index 3.0 (Noelke et al., 2024). We will examine each in turn, justify its inclusion, and discuss what its specification implies for the substantive commitments of the index.

4.1 Median household income at age 35, for children with parents at the 25th percentile

The first criterion variable is the mean rank in the national household income distribution at age 35 of children who grew up in the block group, restricted to children whose parents were at the 25th percentile of the national parental income distribution. The Opportunity Atlas operationalizes this measure by linking decennial census records to longitudinal Internal Revenue Service administrative tax records, using the census records to identify the block group of childhood residence and the IRS records to identify adult earnings (Chetty et al., 2018). The restriction to children of parents at the 25th

percentile of national parental income, rather than averaging across all parental income levels, isolates the upward-mobility component of intergenerational outcomes from the simple replication of parental advantage.

A block group that produces high adult incomes for children born to high-income parents is not, on this measure, distinguished as a vital neighborhood. A block group that produces high adult incomes for children born to lower-income parents is so distinguished. This conceptual commitment follows Chetty and Hendren (2018) in their treatment of intergenerational mobility as the central neighborhood-effect outcome, and it tracks Sampson's (2012) and Sharkey's (2013) treatments of the durable effects of neighborhood conditions across generations.

4.2 Fraction reaching the top quintile of adult household income, across all parental income levels

The second criterion variable is the share of children who grew up in the block group who reach the top quintile of the national household income distribution by age 35, calculated across all parental income levels rather than restricted to a particular parental quintile. This measure is sometimes characterized as an "escape velocity" measure in the mobility literature (Chetty et al., 2014, 2018), in that it captures the share of children for whom the block group's structural conditions did not constrain adult outcomes to the bottom four quintiles of the national distribution. We include it alongside the 25th-percentile-parent measure on the substantive judgment that the two capture related but distinct aspects of neighborhood effect. The first asks whether children of relatively low-income parents do better than parental status would predict, and the second asks whether children's outcomes are bounded by the structural ceiling of the block group regardless of where they begin. We expect the two measures to be highly correlated empirically in the Guilford County data, but their conceptual content differs in ways that justify retaining both as criteria.

4.3 Employment rate at age 35

The third criterion variable is the employment rate at age 35 of adults who grew up in the block group, again derived from the linkage of decennial census childhood records to IRS adult earnings records in the Opportunity Atlas (Chetty et al., 2018). We include employment alongside income on two grounds. First, employment carries information about labor force attachment that income alone does not, and the literature on labor force detachment among prime-age adults in disinvested neighborhoods (Wilson, 1996; Edin & Shaefer, 2015) treats this dimension as substantively distinct from income inadequacy. Second, employment is somewhat less sensitive than income to the cost-of-living adjustments that complicate the interpretation of intergenerational income comparisons across geographies. The three Opportunity Atlas measures together give a more rounded view of intergenerational economic outcomes than any one of them alone would provide.

4.4 Child Opportunity Index 3.0 composite score

The fourth criterion variable is the composite score from the Child Opportunity Index 3.0 (Noelke et al., 2024), which integrates twenty-nine indicators across three domains — education, health and environment, and social and economic — into a single tract-level (with block-group disaggregation in version 3.0) measure of the present-day opportunity structure facing children growing up in the geography. We treat COI as a criterion variable alongside the three Opportunity Atlas measures on the methodological principle that the four criteria together capture both the longitudinal outcome of growing up in a place (the Opportunity Atlas measures, which observe adult outcomes for cohorts of children) and the cross-sectional opportunity structure now in place (the COI measure, which observes contemporary conditions experienced by children currently growing up there). The longitudinal-cross-sectional pairing is methodologically attractive because the two measure adjacent but distinct constructs, and a composite that predicts both well is more credibly capturing structural neighborhood quality than a composite that predicts only one.

The inclusion of COI as a criterion variable raises a methodological concern. The COI 3.0 is itself a composite of twenty-nine indicators, several of which overlap with the predictor variables included in the GCNVI catalog including educational attainment, school proficiency, poverty rate, vacant housing rate, and several others. A predictor that contributes to COI and is also tested for correlation with COI as a criterion will, by construction, correlate with COI. Knighton et al. (2016) raised an analogous concern in the Stanford Reproducible ADI development as criterion variables and predictor variables should not draw from the same data source if construct validity is to be defensible. We address this concern in two ways. First, we will weight the four criteria equally in the screening procedure, so that no single predictor enters the composite solely because it correlates with COI; correlation with at least one of the Opportunity Atlas measures is also required. Second, we conduct a sensitivity analysis in which COI is excluded from the criterion set, the variable selection is repeated against only the three Opportunity Atlas measures, and the resulting variable set is compared to the primary specification. We will report the comparison in the final analytic output.

4.5 Why these four and not others

The literature offers several alternative criterion variables we considered and rejected for this restatement. Life expectancy from the U.S. Small-Area Life Expectancy Estimates Project (CDC NCHS) is available at tract level and would be a defensible criterion under the social-determinants-of-health framing, but it lags neighborhood conditions by approximately a generation and reflects the long-run cumulation of conditions rather than the contemporary structure the index is intended to support. Premature mortality (deaths before age 75) from the CDC National Vital Statistics System has a similar interpretive lag. Educational attainment from the American Community Survey would be defensible but is already widely used as a predictor in the deprivation-index literature and using it as both criterion and predictor would compromise the procedural rigor we are committed to here. Crime rates would be

defensible but their relation to neighborhood vitality is contested in the literature (Sampson, 2012; Sharkey, 2018), and using crime as a criterion would commit the index to a position on that contested relation.

The four criteria we have chosen are the ones we take to be most directly aligned with the substantive purpose of the Atlas (namely, the structural conditions of neighborhoods that condition the long-run economic outcomes of the children who grow up in them) and they are also the four for which the available data permit a methodologically rigorous criterion-validated procedure.

5. Candidate Predictors: Domains and Rationale

We have organized the candidate predictor set into thirteen substantive domains, with approximately one hundred individual measures across the domains. The organization is conceptual and the analytic procedure treats all candidate predictors as a single pool and does not respect domain boundaries in its selection; but the domains structure the variable catalog of Section 6 and clarify the substantive scope of the inquiry. We review the domains briefly here before presenting the catalog itself. We also note that predictor selection is ultimately dependent on acquisition and availability of these datasets.

Domain A covers household economic position. It includes the indicators of income, poverty, public assistance receipt, labor force attachment, and household structure that compose the conventional core of the American deprivation-index tradition (Townsend, 1988; Singh, 2003; EIG, 2025). We include the full set even though several of the indicators are highly correlated with each other; the regression-based refinement with VIF diagnostics will adjudicate among them empirically.

Domain B covers education and human capital and draws on the American Community Survey educational attainment tables, the North Carolina Department of Public Instruction school proficiency records, and the North Carolina Division of Child Development and Early Education licensed-capacity database.

Domain C covers demographic composition including race, ethnicity, age structure, foreign-born status, disability, and household composition and includes the Index of Concentration at the Extremes (Krieger et al., 2016) as a composite measure of racial-economic polarization.

Domain D covers housing conditions as observable in survey data: homeownership, vacancy, tenure, rent and value distributions, cost burden, overcrowding, structural age, and structural condition as reported in the ACS housing tables.

Domain E covers housing and parcel-level administrative records including code violations, tax delinquency, water and electric utility shut-offs by address, foreclosure filings, recent sales activity, and property condition which will require aggregation from parcel to block group and which we treat carefully in the implementation section.

Domain F covers built environment and accessibility, drawing on the Environmental Protection Agency Smart Location Database, the EPA National Walkability Index, the transit GTFS feeds of the Greensboro Transit Authority, the High Point Transit System, and the Piedmont Authority for Regional Transportation.

Domain G covers health and environmental exposure. It draws on CDC PLACES tract-level estimates of chronic disease prevalence (CDC, 2024), the U.S. Small-Area Life Expectancy Estimates Project, the EPA EJScreen environmental justice indicators, and the North Carolina State Center for Health Statistics birth and death records.

Domain H covers the food environment, with measures from the USDA Food Access Research Atlas, commercial supermarket inventories from Data Axle or InfoUSA, and the USDA Food and Nutrition Service SNAP authorized retailer database.

Domain I covers crime and safety, drawing on incident records from the Greensboro Police Department and the High Point Police Department geocoded to block group, and on calls-for-service density as a complementary measure that captures investigated incidents.

Domain J covers civic infrastructure and social capital. As of v5 it is the most substantially expanded domain: an NTEE-decomposed nonprofit typology and organizational-capacity measure built from the IRS Business Master File (arts and culture, human services, youth development, and civic organizations, with reported revenue per capita); voter registration and turnout from the North Carolina State Board of Elections; religious-organization density; the Social Capital Atlas network measures — economic connectedness, clustering, support ratio, volunteering, and civic-organization membership (Chetty et al., 2022a, 2022b); the 2020 Census self-response rate as a behavioral civic-participation proxy; collective efficacy via the PHDCN short scale fielded on Coalition corridor surveys (Sampson, Raudenbush, & Earls, 1997); and library access as physical social infrastructure (Klinenberg, 2018). The expansion answers a structural concern: under criterion-validated selection, a domain that never enters the data can never survive the screen, and the prior four-variable Domain J risked mechanically ‘finding’ that social infrastructure does not matter. Because the project’s central claim is that the nine priority tracts hold assets that conventional deficit-oriented data cannot see, social infrastructure is precisely the asset class the index must measure well (Small, 2009; Klinenberg, 2018).

Domain K is the largest in the catalog and operationalizes the Newkirknomics framework (Newkirk, 2025; Newkirklytics, 2025) that the GCNVI exists to integrate. It includes local business density and revenue capture from the Census County Business Patterns and the Newkirklytics EventureMap vendor registry; entrepreneur pipeline measures from the Census Nonemployer Statistics and the Genesis Bridge intake; capital access measures from the Federal Financial Institutions Examination Council Home Mortgage Disclosure Act (HMDA) database, the Small Business Administration loan records, and the CDFI Fund Transaction-Level Reports; anchor procurement engagement; economic

infrastructure capacity from incubator and accelerator program rosters; and commercial corridor activity from corridor-specific occupancy and sales surveys.

Domain L covers consumer spending and market characteristics, including ACS-derived disposable income estimates, retail leakage analyses, and FDIC Summary of Deposits records aggregated by branch.

Domain M covers digital infrastructure, including broadband subscription from the American Community Survey and FCC Form 477 deployment data.

Domain N comprises the existing external composite indices that we include as candidate predictors in their own right: the Area Deprivation Index national rank, the Social Deprivation Index score, the CDC/ATSDR Social Vulnerability Index percentile, the Economic Innovation Group Distressed Communities Index quintile, the U.S. Census Bureau Multidimensional Deprivation Index, and the Climate and Economic Justice Screening Tool designation.

The catalog is a candidate set, not a final variable list. Some candidate variables will fail the Pearson correlation screen and will be excluded; others will fail the variance inflation factor diagnostic and will be excluded; still others will be retained as part of the final summative composite. Other candidate predictors may be suggested by the Coalition of Anchors. The catalog's purpose is to make the substantive scope of inquiry visible to the Coalition of Anchors and Guilford County partners and to provide the data-source documentation necessary for the analytic team to assemble the inputs.

6. Variable Catalog

The catalog below presents approximately one hundred candidate predictors organized into the thirteen domains introduced in Section 5, plus the four criterion variables of Section 4 displayed first for reference. For each variable we specify the operational definition or formula, the primary data source (with secondary sources noted where applicable), and the rationale for inclusion, citing the literature on which the rationale rests. All variables are reported at the block-group level where data are available; where data are available only at the tract level, tract values are assigned to all constituent block groups and the variable is noted as tract-derived. Direction of expected correlation with the four criterion variables (positive or negative) is not displayed in the catalog because the analytic procedure determines the direction empirically rather than imposing it a priori; for variables whose direction is conventionally understood (e.g., poverty rate, vacancy rate), the empirical direction will simply confirm the conventional one and the variable will enter the composite with sign appropriate to the correlation.

6.1 Criterion Variables (Y_1 – Y_4)

These four variables anchor the criterion-validated variable selection. They are not included in the summative composite; they are the targets against which candidate predictors are screened.

Variable	Operationalization	Data Source	Rationale & Citation
Y1 Adult HH Income (parents @25th pctl)	Mean household income rank at age 35 for children with parents at the 25th percentile of the national income distribution who grew up in the BG.	Opportunity Atlas v2 (Chetty et al., 2018, 2026).	Isolates upward-mobility component of intergenerational outcomes from replication of parental advantage (Chetty & Hendren, 2018; Sharkey, 2013).
Y2 Fraction Reaching Top Quintile	Share of children who grew up in BG reaching the top 20% of national adult household income by age 35, all parental income levels.	Opportunity Atlas v2 (Chetty et al., 2018, 2026).	Captures structural ceiling of BG on long-run outcomes; complements Y1 by addressing escape velocity (Chetty et al., 2014).
Y3 Adult Employment Rate	Employment rate at age 35 for adults who grew up in the BG.	Opportunity Atlas v2 (Chetty et al., 2018, 2026).	Adds labor force attachment dimension distinct from income (Wilson, 1996; Edin & Shaefer, 2015).
Y4 Child Opportunity Index 3.0	Composite of 29 indicators across education, health/environment, social/economic; BG-level in v3.0.	diversitydatakids.org / Brandeis Institute for Child, Youth and Family Policy (Noelke et al., 2024).	Cross-sectional present-day opportunity structure complements longitudinal mobility outcomes (Acevedo-Garcia et al., 2020).

6.2 Domain A. Household Economic Position

Income, poverty, public assistance receipt, labor force attachment, household structure. The conventional core of the American deprivation-index tradition (Townsend, 1988; Singh, 2003; Butler et al., 2013; EIG, 2025).

Variable	Operationalization	Data Source	Rationale & Citation
A1 Median Household Income	Tract median household income; reported in 2022 inflation-adjusted dollars.	ACS 5-Year 2019–2023, Table B19013.	Core variable in DCI, ADI, SDI. Direct measure of household economic position (Singh, 2003).
A2 Median HH Income — County Ratio	Tract median HH income / Guilford County median HH income.	ACS Table B19013, computed.	Adopted by DCI to enable cross-tract comparability under varying regional cost structures (EIG, 2025).
A3 Poverty Rate (All Ages)	Share of persons with income below the federal poverty threshold.	ACS 5-Year 2019–2023, Table B17001.	Universal in deprivation indices; Sills (2022) used in FHHP Health Equity Score.
A4 Deep Poverty Rate (<50% FPL)	Share of persons with income below 50% of the federal poverty threshold.	ACS 5-Year 2019–2023, Table B17002.	Sharkey (2013) treats deep poverty as substantively distinct from poverty at the 100% threshold.

Variable	Operationalization	Data Source	Rationale & Citation
A5 Child Poverty Rate	Share of persons under 18 with family income below 100% FPL.	ACS 5-Year 2019–2023, Table B17006.	Foregrounded by COI 3.0 and by the next-generation framing of Sharkey (2013) and Chetty & Hendren (2018).
A6 SNAP Receipt	Share of households receiving SNAP in past 12 months.	ACS 5-Year 2019–2023, Table B22003.	Sills (2022) used SNAP receipt as most reliable tract-level food-insecurity indicator.
A7 Public Assistance Income	Share of households with cash public assistance income in past 12 months.	ACS 5-Year 2019–2023, Table B19057.	Used in CHCS 2015 Market Segmentation (Sills, 2015); captures non-SNAP means-tested participation.
A8 Prime-Age Employment Ratio	Share of population 25–54 currently employed (employed-to-population ratio).	ACS 5-Year 2019–2023, Tables B23001, B23002.	Adopted by DCI; captures labor force detachment rather than active-search unemployment (EIG, 2025).
A9 Unemployment Rate	Share of civilian labor force aged 16+ unemployed.	ACS 5-Year 2019–2023, Table B23025.	Conventional measure; included for comparison with A8 (Tabachnick & Fidell, 2019).
A10 Not in Labor Force (Age 25–64)	Share of population 25–64 not in the labor force.	ACS 5-Year 2019–2023, Table B23001.	Sills (2022) Health Equity Score negative variable; captures Wilson (1996) labor-detachment dimension.
A11 Single-Parent Households w/ Children	Share of family households with own children under 18 headed by single parent.	ACS 5-Year 2019–2023, Table B11003.	Included in COI (Noelke et al., 2024); proxy for household economic resource constraint.
A12 Households with No Vehicle	Share of occupied housing units with no vehicle available.	ACS 5-Year 2019–2023, Table B25044.	Used in SVI (Flanagan et al., 2011) and Sills (2015) High Point Market Segmentation.
A13 Gini Coefficient (Income Inequality)	Tract-level Gini coefficient of household income; BG values imputed from tract.	ACS 5-Year 2019–2023, Table B19083 (tract).	Within-tract inequality complements between-tract inequality (Krieger et al., 2017).

6.3 Domain B. Education and Human Capital

Educational attainment, school proficiency, early-childhood capacity. The forward-looking dimension foregrounded by COI 3.0 (Noelke et al., 2024) and by the Chetty et al. neighborhood-effect literature.

Variable	Operationalization	Data Source	Rationale & Citation
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Variable	Operationalization	Data Source	Rationale & Citation
B1 Less Than HS Attainment	Share of adults 25+ with no high school diploma or equivalent.	ACS 5-Year 2019–2023, Table B15003.	Used in ADI, SDI, DCI, SVI; threshold below which long-run labor market participation is constrained (Singh, 2003).
B2 HS Equivalent or Higher	Share of adults 25+ with HS diploma or equivalent.	ACS 5-Year 2019–2023, Table B15003.	Conventional positive deprivation-index variable; Sills (2022) Health Equity Score.
B3 Bachelor’s Degree or Higher	Share of adults 25+ with BA or higher.	ACS 5-Year 2019–2023, Table B15003.	Captures dimension most strongly associated with intergenerational mobility in Chetty et al. (2014).
B4 Reading Proficiency (3rd-8th grade)	Weighted average % proficient on NC EOG reading assessment, schools serving BG.	NC DPI School Report Cards + GCS attendance-zone-to-BG crosswalk.	COI 3.0 educational subdomain; FHHP/GJ30 work documented school quality as top resident concern (Sills, 2022).
B5 Math Proficiency (3rd-8th grade)	Weighted average % proficient on NC EOG math assessment, schools serving BG.	NC DPI School Report Cards + GCS attendance-zone-to-BG crosswalk.	Complements B4; treated separately in COI (Noelke et al., 2024).
B6 Preschool Enrollment (Age 3–4)	Share of children age 3–4 enrolled in school (pre-K or kindergarten).	ACS 5-Year 2019–2023, Table B14002.	Included in COI early-childhood subdomain; early-childhood inputs predict later attainment (Heckman, 2006).
B7 Licensed Childcare Capacity	Licensed childcare seats per 100 children under 6.	NC DCDEE licensed facility database, geocoded.	Sills (2024) Food Study documented childcare access as binding constraint on parental labor force participation.
B8 Limited English Speaking HH	Share of households with no member 14+ speaking English well.	ACS 5-Year 2019–2023, Table B16002.	SVI Household Composition theme; captures linguistic barriers to service access (Flanagan et al., 2011).

6.4 Domain C. Demographic Composition

Racial, ethnic, age, foreign-born, disability composition. Treated cautiously: composition variables can produce misleading results when used as deprivation proxies (Feldman et al., 2015), so we include the Index of Concentration at the Extremes (Krieger et al., 2016) as a structural-position composite alongside the raw composition measures.

Variable	Operationalization	Data Source	Rationale & Citation
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Variable	Operationalization	Data Source	Rationale & Citation
C1 % Black or African American	Share of population identifying as Black or African American alone.	ACS 5-Year 2019–2023, Table B02001.	Included as raw composition for descriptive purposes; structural interpretation requires C5 ICE Race-Income (Krieger et al., 2016).
C2 % Hispanic or Latino	Share of population identifying as Hispanic/Latino of any race.	ACS 5-Year 2019–2023, Table B03003.	As C1; included for descriptive completeness.
C3 % Foreign-Born	Share of population foreign-born.	ACS 5-Year 2019–2023, Table B05002.	SVI Household Composition theme; immigrant settlement patterns interact with structural conditions (Flanagan et al., 2011).
C4 Age Dependency Ratio	(Population under 18 + Population 65+) / Population 18–64.	ACS 5-Year 2019–2023, Table B01001, computed.	Conventional measure of demographic dependency burden; included in SVI.
C5 ICE Race-Income	(White HH with income $\geq$ \$100K – Black/Hispanic HH with income $<$ \$25K) / Total HH; range –1 to +1.	ACS 5-Year 2019–2023, Tables B19001A–I, computed.	Krieger et al. (2016) measure of racialized economic position; complements raw composition variables (Feldman et al., 2015).
C6 % With Disability	Share of civilian noninstitutionalized population with one or more disabilities.	ACS 5-Year 2019–2023, Table B18101.	SVI Household Composition theme; included in Sills (2022) Health Equity Score.
C7 Population Density	Population per square mile in BG land area.	ACS B01003 / TIGER/Line BG polygons.	Controls for built-environment density; relevant to interpretation of business and amenity counts.

6.5 Domain D. Housing Conditions

Housing tenure, vacancy, value, cost burden, age, and quality as observable in the American Community Survey. Central to the Mallach (2008) market segmentation tradition and to the Townsend (1988) deprivation framework.

Variable	Operationalization	Data Source	Rationale & Citation
D1 Homeownership Rate	Share of occupied housing units that are owner-occupied.	ACS 5-Year 2019–2023, Table B25003.	Genesis Block Pillar 3 (Ownership Economy); primary vehicle of intergenerational wealth in US (Rothstein, 2017; Faber, 2020).

Variable	Operationalization	Data Source	Rationale & Citation
D2 Housing Vacancy Rate	Share of housing units vacant (incl. seasonal exclusion).	ACS 5-Year 2019–2023, Table B25002.	DCI core indicator; Mallach (2008, 2012) treats vacancy as the central market-segmentation signal.
D3 Vacant for Sale (Not Seasonal)	Share of housing units vacant and for sale.	ACS 5-Year 2019–2023, Table B25004.	Refines D2 to active market vacancy; distinguishes turnover from abandonment (Mallach, 2018).
D4 Renter-Occupied Share	Share of occupied housing units that are renter-occupied.	ACS Table B25003, computed.	Used in CHCS 2015 Market Segmentation (Sills, 2015); high renter share signals different market dynamics.
D5 Median Home Value	Tract median value of owner-occupied housing units.	ACS 5-Year 2019–2023, Table B25077.	Wealth holding for owners; the Petterson (2023) ADI critique is honored through pre-z-score standardization.
D6 Median Gross Rent	Tract median gross rent paid by renter-occupied units.	ACS 5-Year 2019–2023, Table B25064.	Captures rental market price; complements D5.
D7 Renter Cost Burden ≥30%	Share of renter HH paying ≥30% of income on gross rent.	ACS 5-Year 2019–2023, Table B25070.	Established by JCHS (2024) as most defensible single measure of housing-cost stress for renters.
D8 Severe Renter Cost Burden ≥50%	Share of renter HH paying ≥50% of income on gross rent.	ACS 5-Year 2019–2023, Table B25070.	Captures severe affordability stress; tracks displacement risk (JCHS, 2024).
D9 Owner Cost Burden ≥30%	Share of owner HH with SMOC ≥30% of income.	ACS 5-Year 2019–2023, Table B25091.	Owner-side complement to D7; relevant in older paid-off owner stock common in target tracts.
D10 Overcrowding (>1.5 per room)	Share of occupied units with more than 1.5 occupants per room.	ACS 5-Year 2019–2023, Table B25014.	SVI Housing Type theme; captures household-level space inadequacy.
D11 Pre-1950 Housing Stock	Share of housing units built before 1950.	ACS 5-Year 2019–2023, Table B25034.	Proxy for lead-paint exposure risk and deferred-maintenance burden; HOLC-era marker.
D12 Mobile Home Share	Share of housing units classified as mobile homes.	ACS 5-Year 2019–2023, Table B25024.	SVI Housing Type theme; relevant in Guilford County exurban BG.
D13 Lacking Complete Plumbing	Share of occupied units lacking complete plumbing facilities.	ACS 5-Year 2019–2023, Table B25049.	Townsend (1988) housing-deprivation tradition; rare but substantively important when present.

Variable	Operationalization	Data Source	Rationale & Citation
D14 Moved in Past Year	Share of population that moved within the past 12 months (transience).	ACS 5-Year 2019–2023, Table B07003.	Sampson (2012) residential stability literature; high transience disrupts collective efficacy.
D15 SF Detached / MF Ratio	Single-family detached units / all multi-family units (≥ 2 -unit structures).	ACS 5-Year 2019–2023, Table B25024, computed.	Mallach (2018) treats SF/MF mix as a structural market segmentation variable.

6.6 Domain E. Housing and Parcel-Level Administrative Records

Parcel-level conditions aggregated to block group. The most operationally important domain for the Coalition of Anchors land-intelligence layer, and the domain in which the GCNVI will provide the most decision-relevant detail beyond what the deprivation-index tradition can offer. Requires careful parcel-to-BG aggregation rules specified in Section 7.

Variable	Operationalization	Data Source	Rationale & Citation
E1 Code Violations Density	Annual code-enforcement actions per 1,000 housing units, 3-year rolling average.	City of Greensboro Code Compliance; City of High Point Community Development.	Used in CHCS 2015 (Sills, 2015); High Point uses proactively for blight prevention.
E2 Tax-Delinquent Parcels	Share of parcels with delinquent property taxes (current year + prior years).	Guilford County Tax Office tax delinquency records.	Mallach (2008) treats delinquency as leading indicator of abandonment; Sills (2015) High Point.
E3 Water Service Shut-Offs by Address	Annual water service disconnections per 1,000 service addresses.	Greensboro Water Resources; High Point Public Services.	Direct indicator of household financial distress; complements ACS poverty measures.
E4 Vacant / Abandoned Parcels	Share of parcels coded vacant or abandoned in county records.	Guilford County Parcel Records; Tax Office.	Mallach (2012, 2018) treats abandonment as central market signal distinct from ACS vacancy.
E5 Investor-Owned SF Share	Share of single-family detached parcels owned by LLC, LP, REIT, or trust with multi-property holdings.	Guilford ROD parcel + ownership; NC SOS entity registry; UNC Charlotte methodology.	Direct linkage to Project 2 of IRI-Thigpen engagement; Raymond et al. (2018) on investor effects.
E6 Foreclosure Filings Density	Foreclosure filings per 1,000 owner-occupied units, 3-year rolling average.	Guilford County Clerk of Court foreclosure docket.	Mallach (2018) leading market-distress indicator.
E7 Recent Sales Velocity	Single-family sales per 100 owner units, 12-month trailing.	Guilford County ROD deed records.	Market liquidity signal; complements vacancy in Mallach (2008) typology.

Variable	Operationalization	Data Source	Rationale & Citation
E8 Property Condition Rating	Share of parcels rated below “Average” in county assessor records.	Guilford County Tax Assessor property condition codes.	Sills (2015) High Point Market Segmentation used assessor condition ratings.
E9 Recent Permit Activity	Building permits issued per 100 housing units, 12-month trailing.	City of Greensboro Inspections; City of High Point Permits.	Positive market signal; Mallach (2018) treats permit volume as investment indicator.
Variable	Operationalization	Data Source	Rationale & Citation
E10 Eviction Filing Rate	Eviction filings per 100 renter households; 2010-tract series crosswalked to 2020 BGs.	Eviction Lab, Princeton University (host allowlist addition pending; manual download otherwise).	Desmond (2016) on forced displacement and the destruction of neighborhood ties; complements E6 foreclosures on the rental side.

6.7 Domain F. Built Environment and Accessibility

Walkability, transit access, job accessibility, parks, land use mix. Operationalizes the COI 3.0 built-environment subdomain and the EPA Smart Location Database framework.

Variable	Operationalization	Data Source	Rationale & Citation
F1 Walkability Index	EPA National Walkability Index composite score (0–20), averaged across BG.	EPA Smart Location Database v3.0.	COI 3.0 walkability subdomain; complements F2 transit access (Noelke et al., 2024).
F2 Transit Access (½ Mile)	Share of BG population within ½ mile of a transit stop with ≥4 weekday peak trips/hour.	GTA, HEAT, PART GTFS feeds; computed via network buffer.	Sills (2025) Washington Street finding on transit as resident-identified constraint.
F3 Job Accessibility — 45-min Auto	Total employment accessible within 45-min auto travel time.	EPA Smart Location Database, D5br variable.	COI economic-opportunity subdomain; central in commute-shed analysis (EPA, 2021).
F4 Job Accessibility — 45-min Transit	Total employment accessible within 45-min transit travel time.	EPA Smart Location Database, D5cri variable.	Captures equity of job access for transit-dependent populations.
F5 Park Access (½ Mile)	Share of BG population within ½ mile of public park or open space.	PAD-US, City of Greensboro/High Point parks layers.	COI health-and-environment subdomain; relevant for child outdoor activity.
F6 Tree Canopy Coverage	Share of BG land area with tree canopy cover.	USFS Tree Canopy Cover; NLCD 2021.	Heat-island and air-quality mediation (Schwarz et al., 2015).

Variable	Operationalization	Data Source	Rationale & Citation
F7 Land Use Mix Entropy	Shannon entropy of land use category proportions in BG.	EPA Smart Location Database, D2b_e8mix variable.	Captures mixed-use urbanism; correlates with walkability and economic activity.
F8 Impervious Surface Share	Share of BG land area covered by impervious surface.	NLCD 2021 Impervious Surface.	Built-up density proxy; heat-island and stormwater risk.

6.8 Domain G. Health and Environmental Exposure

Chronic disease prevalence, life expectancy, environmental hazards exposure. Drawn from CDC PLACES (CDC, 2024), USALEEP, EPA EJScreen, and NC State Center for Health Statistics. Most variables are tract-level and will be assigned to constituent block groups.

Variable	Operationalization	Data Source	Rationale & Citation
G1 Life Expectancy at Birth	Tract-level estimated life expectancy at birth; BG values from tract.	CDC NCHS USALEEP 2010–2015 (most recent).	Most defensible single summary indicator of population health; Sills (2022) Health Equity Score.
G2 Adult Diabetes Prevalence	Model-based estimate of adult diabetes diagnosis prevalence.	CDC PLACES 2024 release.	Representative chronic disease most strongly associated with social determinants (CDC, 2024).
G3 Hypertension Prevalence	Model-based estimate of adult hypertension prevalence.	CDC PLACES 2024 release.	Sills (2022) FHHP variable set; complements G2 in chronic-disease panel.
G4 Obesity Prevalence	Model-based estimate of adult obesity prevalence.	CDC PLACES 2024 release.	Sills (2022) FHHP variable set; metabolic-syndrome marker.
G5 Mental Health Distress	Model-based estimate of adults with ≥ 14 days of poor mental health in past 30.	CDC PLACES 2024 release.	FHHP, Healthy Steps, and Washington Street studies (Sills, 2022, 2025) document mental health as top concern.
G6 No Health Insurance	Model-based estimate of adults 18–64 without health insurance.	CDC PLACES 2024 + ACS B27001.	Sills (2022) Health Equity Score; access-to-care barrier.
G7 Premature Mortality (Age-Adjusted)	Age-adjusted years of potential life lost before age 75 per 100,000.	NC State Center for Health Statistics death records, 2018–2022 aggregated.	Conventional health-disparity outcome; CHR (UWPHI, 2014) length-of-life construct.
G8 Low Birthweight	Share of live births under 2,500g.	NC State Center for Health Statistics birth records, 2018–2022.	Early-life indicator of maternal-and-child stress; FHHP focus.

Variable	Operationalization	Data Source	Rationale & Citation
G9 EJScreen Pollution Burden	Composite percentile across PM2.5, ozone, diesel PM, air toxics cancer risk, lead paint, traffic proximity.	EPA EJScreen 2024 release.	CEJST-adjacent multi-pollutant exposure; aligns with environmental-justice framing (CEQ, 2022).
G10 Lead Paint Risk	EJScreen lead-paint indicator (function of pre-1960 housing share).	EPA EJScreen 2024; ACS B25034.	Persistent neurodevelopmental hazard for children; particularly relevant to Y4 COI.

6.9 Domain H. Food Environment

Supermarket access, food retail diversity, SNAP retailer access. Directly tied to the Guilford County Food Study (Sills, 2024) and to the Foundation for a Healthy High Point Food Council priorities.

Variable	Operationalization	Data Source	Rationale & Citation
H1 USDA Low-Income Low-Access	Binary: BG meets USDA “food desert” criteria (LI + ≥1 mile to supermarket urban).	USDA ERS Food Access Research Atlas, 2019.	Sills (2024) Guilford County Food Study; CHCS work; CHR food-environment construct.
H2 Supermarket Density	NAICS 44511 establishments per 10K residents.	Census County Business Patterns; Data Axle Business Database.	Sills (2024) Food Study supermarket inventory; conventional access measure.
H3 Convenience-Heavy Retail Ratio	Convenience stores (NAICS 44512) ÷ supermarkets (NAICS 44511) in BG + ½-mile buffer.	Data Axle; Census CBP.	Captures retail-mix imbalance documented in Healthy High Point Food Council priorities.
H4 SNAP-Authorized Retailer Density	SNAP-authorized retailers per 10K residents.	USDA FNS SNAP Retailer Locator.	Captures functional access to food retail for SNAP-receiving households.

6.10 Domain I. Crime and Safety

Part 1 offenses and calls-for-service density. The relation of crime to neighborhood vitality is contested in the literature (Sampson, 2012; Sharkey, 2018), and we include crime indicators as candidate predictors without committing the index to a position on causal direction.

v5 sourcing note: I1–I5 are populated on an interim basis from the three-agency build (Greensboro Police Department, High Point Police Department, and the Guilford County Sheriff’s Department, 2014–2018; 2010-block-group native) produced for the Guilford County Substance Use study (Sills, 2020) and published by the UNCG Department of Geography, Environment & Sustainability; source data were supplied directly by the three agencies. A current-vintage incident build (GPD open-data FeatureServer plus an HPPD records request) will refresh the series. The 2014–2018 window is flagged in all outputs.

Variable	Operationalization	Data Source	Rationale & Citation
11 Violent Crime Rate	Part 1 violent offenses per 1,000 residents, 3-year rolling average.	Greensboro Police Dept; High Point Police Dept; geocoded to BG.	Sampson, Raudenbush & Earls (1997) collective-efficacy framework; FBI UCR convention.
12 Property Crime Rate	Part 1 property offenses per 1,000 residents, 3-year rolling average.	Greensboro PD; High Point PD; geocoded to BG.	Conventional measure; included in CHCS 2015 (Sills, 2015) High Point Market Segmentation.
13 Calls-for-Service Density	All police calls for service per 1,000 residents, 3-year rolling average.	Greensboro PD CAD; High Point PD CAD.	Captures investigated incidents that did not result in Part 1 classification (Sharkey, 2018).
14 Firearm-Involved Incidents	Firearm-involved Part 1 incidents per 1,000 residents.	Greensboro PD; High Point PD.	Krieger et al. (2017) on ICE Race-Income and firearm violence; relevant to long-run child outcomes.
15 Domestic Violence Calls	Domestic-violence-coded calls per 1,000 residents.	Greensboro PD CAD; High Point PD CAD.	Sharkey (2013) on household exposure as neighborhood-mediated outcome.

Variable	Operationalization	Data Source	Rationale & Citation
16 AGS CrimeRisk Indexes (Cross-Validation)	AGS 2022 total, personal, and property crime indexes (US average = 100); 2020-tract values broadcast to BGs.	Applied Geographic Solutions via Esri; interim pull from a public ArcGIS re-share — PROVISIONAL; licensed pull required before publication.	Modeled from FBI UCR plus ~100 socioeconomic predictors (AGS, 2024); cross-validation for the recorded-incident series only, given endogeneity with index covariates and limited independent validation.
17 Neighborhood Incarceration Rate (Childhood Cohort)	Share of children (parents at the 25th percentile) who grew up in the tract and were incarcerated on 2010 Census day; crosswalked to 2020 BGs.	Opportunity Atlas full tract outcomes (jail_pooled_pooled_p25).	Sampson & Loeffler (2010) on the local concentration of mass incarceration; a justice-involvement construct distinct from contemporaneous crime, at zero marginal acquisition cost.

6.11 Domain J. Civic Infrastructure and Social Capital

Expanded in v5 from four variables to eleven. Operationalizes the Putnam (2000) social-capital tradition, the Klinenberg (2018) social-infrastructure argument, the Chetty et al. (2022a, 2022b) economic-connectedness findings, and the COI 3.0 social-and-economic subdomain.

Variable	Operationalization	Data Source	Rationale & Citation
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Variable	Operationalization	Data Source	Rationale & Citation
J1 Nonprofit Organization Density	501(c)(3) public charities located in BG per 10,000 residents.	IRS Business Master File via NCCS; UWGG + HPCF grantee rosters.	Putnam (2000); COI 3.0 civic-infrastructure proxy (Noelke et al., 2024).
J2 Voter Registration Rate	Registered voters as share of citizen voting-age population.	NC State Board of Elections; ACS B05003.	Civic engagement measure; Putnam (2000) tradition.
J3 Voter Turnout (General)	Turnout in most recent general election as share of registered voters.	NC State Board of Elections.	Active civic engagement; complements J2.
J4 Religious Organization Density	Religious establishments (NAICS 8131) per 10,000 residents.	Census CBP; Data Axle.	Putnam (2000); historically central social-capital site in NC African American communities.

Variable	Operationalization	Data Source	Rationale & Citation
J1a–J1e NTEE-Decomposed Nonprofit Density & Capacity	J1 refined into arts & culture (NTEE A), human services (I–P), youth development (O), and civic/community improvement (R–W) organizations per 10,000 residents, plus BMF-reported revenue per capita (capacity proxy; 990 program-expense build to follow). Addresses batch-geocoded to 2020 BG.	IRS Business Master File via ProPublica Nonprofit Explorer API; U.S. Census Bureau batch geocoder.	Organizational typology matters more than counts (Small, 2009; Klinenberg, 2018); arts organizations were previously absent from the catalog. CAVEAT: organizational address ≠ service area (Allard, 2009).
J5 Economic Connectedness	Share of above-median-SES friends among below-median-SES residents; ZIP-level, ZCTA→tract land-area allocation, tract→BG broadcast.	Meta/Opportunity Insights Social Capital Atlas (Chetty et al., 2022a, 2022b); local project file.	The strongest neighborhood-level correlate of upward mobility in the source analyses, measured independently (Facebook networks) of the tax-record criteria — informative, not circular. Area-weighted allocation flagged.
J6 Network Cohesion (Clustering; Support Ratio)	Friendship-network clustering coefficient; share of friendships with a mutual friend. ZIP→tract→BG as J5.	Social Capital Atlas.	Bonding capital distinct from bridging (Coleman, 1988); complements J5.
J7 Civic Engagement (Volunteering; Civic Organizations)	Volunteering rate; civic-organization membership density. ZIP→tract→BG as J5.	Social Capital Atlas.	Behavioral engagement complementing administrative J2/J3 (Putnam, 2000).
J8 Census Self-Response Rate	2020 Decennial cumulative (CRRALL) and internet (CRRINT) self-response rates; tract-level, broadcast to BGs.	U.S. Census Bureau 2020 response-rate API.	Behavioral proxy for civic participation and institutional trust; no geocoding required; used in the Bureau's hard-to-count literature.

Variable	Operationalization	Data Source	Rationale & Citation
J9 Collective Efficacy (Survey)	5-item PHDCN short scale (social cohesion + informal social control) on Coalition corridor surveys; BG aggregation with small-area reliability assessment.	Partner-collected with K11–K13.	The canonical social-process construct (Sampson, Raudenbush, & Earls, 1997); no administrative source exists.
J10 Library Access (½ Mile)	Share of BG population within ½ mile of a public library outlet (F5 pattern).	IMLS Public Libraries Survey outlet file (manual).	Klinenberg's (2018) flagship social-infrastructure institution.

6.12 Domain K. Economic/Business Activity Newkirknomics Framework

The eleven indicators of the Newkirknomics NEI/NBF framework (Newkirk, 2025; Newkirklytics, 2025), supplemented by NC Commerce LEED records and federal lending databases. The most operationally novel domain in the catalog and the principal locus of integration with the Coalition of Anchors Pilot.

Variable	Operationalization	Data Source	Rationale & Citation
K1 Local Business Density	Locally owned establishments per 10,000 residents (Newkirklytics LBD).	Census CBP 2022; EventureMap vendor registry; NC Commerce LEED.	Newkirknomics Pillar 1; structural precondition for capital absorption (Newkirk, 2025).
K2 Establishments by Size — Small	Establishments with <20 employees per 10K residents.	Census CBP 2022.	Captures small-firm density distinct from total; relevant to local-ownership thesis.
K3 Black-Owned Business Density	Establishments classified Black-owned per 10K residents (firms + nonemployers).	Census Annual Business Survey 2021; supplemented by ABS suppression handling.	Genesis Block Pillar 3 (Ownership Economy); Newkirklytics tracking.
K4 Women-Owned Business Density	Establishments classified woman-owned per 10K residents.	Census ABS 2021.	Diversity-of-ownership measure; complements K3.
K5 Nonemployer Firm Density	Sole proprietors and freelance firms per 10K residents.	Census Nonemployer Statistics 2021.	Newkirknomics Entrepreneur Pipeline Strength component (Newkirklytics, 2025).
K6 New Business Formation Rate	New revenue-generating firms per year per 10K residents (YoY delta).	Census Nonemployer Statistics 2020–2021 deltas; NC SOS Business Registration.	DCI establishment-change variable; Newkirknomics formation component.
K7 HMDA Mortgage Originations	Mortgage originations per 100 owner-occupied units, 3-year average.	FFIEC HMDA 2021–2023.	Capital access; redlining literature on durable credit-access constraints (Faber, 2020).

Variable	Operationalization	Data Source	Rationale & Citation
K8 SBA Small Business Lending	SBA 7(a) + 504 loans per 1,000 establishments, 3-year average.	SBA Loan Data Reports.	Newkirknomics Capital Access pillar (Newkirklytics, 2025).
K9 CDFI Loan Activity	CDFI Fund award recipient originations per 10K residents.	CDFI Fund Transaction-Level Reports; Self-Help, Truliant filings.	Mission-aligned capital essential in disinvested neighborhoods (Faber, 2020).
K10 CRA Small Business Lending	CRA-credited small business loans per 1,000 establishments.	FFIEC CRA Aggregate Data.	Bank reinvestment in low- to moderate-income areas; CRA performance evaluation.
K11 Anchor Procurement Capture	\$ local procurement directed to BG-resident-serving firms per 10K residents.	Coalition of Anchors self-reporting; Bennett, NC A&T, Cone Health MOU tracking.	Newkirknomics APE pillar; Coalition theory of change (Newkirk, 2025).
K12 Incubator/Accelerator Seats	Verified annual business support program seats per 10K residents.	SBTDC NC A&T; Greensboro Chamber; BTW Society; Genesis Block ILM.	Newkirknomics Economic Infrastructure Capacity (Newkirklytics, 2025).
K13 Commercial Corridor Occupancy	Share of ground-floor commercial space occupied in active corridors.	Anchor Coalition corridor surveys; DGI; Washington Street District inventory.	Newkirknomics CCA pillar; visible signal of corridor health (Sills, 2025).

6.13 Domain L. Consumer Spending and Market Activity

Disposable income, retail leakage, bank deposits. Captures the demand-side complement to the production-side measures of Domain K.

Variable	Operationalization	Data Source	Rationale & Citation
L1 Estimated Consumer Expenditure	Modeled total consumer expenditure per HH, from BLS CES regional patterns × ACS income distribution.	BLS Consumer Expenditure Survey; ACS B19001.	Demand-side capacity in neighborhood economy (Newkirklytics, 2025).
L2 Retail Leakage Ratio	Estimated resident retail spending capacity ÷ actual retail sales in BG.	Esri Business Analyst or Newkirklytics modeled estimate.	Newkirknomics Local Spending Retention (ECM) component (Newkirk, 2025).
L3 Bank Deposits per Capita	Bank branch deposits allocated to BG-resident customers per capita.	FDIC Summary of Deposits; modeled BG allocation.	Local wealth holding measure; complements K7 origination data.

6.14 Domain M. Digital Infrastructure

Broadband subscription, device access. Operationalizes Newkirknomics Pillar 4 (Digital Access is Fundamental; Newkirk, 2025) and the Piedmont Triad Digital Inclusion Plan (Sills, 2024).

Variable	Operationalization	Data Source	Rationale & Citation
M1 Broadband Subscription	Share of HH with broadband internet subscription.	ACS 5-Year 2019–2023, Table B28011.	Newkirknomics Pillar 4; COI 3.0 social-economic subdomain (Noelke et al., 2024).
M2 Computer Ownership	Share of HH with one or more computing devices.	ACS 5-Year 2019–2023, Table B28001.	Sills (2022) FHHP Health Equity Score; complements M1.
M3 FCC Broadband Deployment	Maximum advertised download speed available to BG.	FCC Form 477 / National Broadband Map.	Supply-side complement to ACS subscription; deployment vs. adoption distinct (NTIA, 2022).
M4 Mobile-Only Internet Households	Share of HH with smartphone-only internet (no wireline broadband).	ACS 5-Year 2019–2023, Table B28010.	Captures functional digital exclusion (Sills, 2024).

6.15 Domain N. Existing External Composites — Candidate Predictors and Triangulation Comparators

Established neighborhood indices included both as candidate predictors in the analytic procedure and as triangulation comparators in the final tract scorecards. Their inclusion as predictors tests whether the structural information they capture survives the criterion-validated screening when finer-grained component measures are also available.

Variable	Operationalization	Data Source	Rationale & Citation
N1 Area Deprivation Index	ADI national rank (1–100); tract-level assigned to BG.	University of Wisconsin Neighborhood Atlas; ADI v3.2.	Singh (2003); Kind & Buckingham (2018); Petterson (2023) standardization critique observed.
N2 Social Deprivation Index	SDI score (0–100), tract-level.	Robert Graham Center.	Butler et al. (2013); health-services research convention.
N3 Social Vulnerability Index	Overall SVI percentile (0–1); tract-level assigned to BG.	CDC/ATSDR 2022 release.	Flanagan et al. (2011); disaster-management framing but multi-purpose vulnerability composite.
N4 Distressed Communities Index	DCI quintile (1–5); ZIP-level mapped to BG via crosswalk.	Economic Innovation Group, 2025.	Foregrounds economic and labor-market dimensions; closest peer to GCNVI (EIG, 2025).
N5 Multidimensional Deprivation Index	MDI score (count of deprivations 0–6); tract-level assigned to BG.	U.S. Census Bureau; Glassman (2019, 2021).	Sen-inflected capabilities approach (Alkire & Foster, 2011); deprivation joint distribution.

Variable	Operationalization	Data Source	Rationale & Citation
N6 CEJST Disadvantaged Status	Binary: CEJST disadvantaged tract designation.	CEQ Climate and Economic Justice Screening Tool, 2022.	Federal funding eligibility marker (CEQ, 2022).

7. Analytic Procedure

This section sets out the analytic procedure for the GCNVI in operational detail. The procedure has six sequential steps and several parallel diagnostic and sensitivity-analytic checks. We describe each step in terms specific enough that a subsequent analyst could implement the procedure in SPSS or in R without further instruction. We anchor the description to the SPSS environment because we specified visual binning as part of the procedure and the SPSS Visual Binning tool implements the Mean $\pm$ SD cut points natively.

7.1 Step 1 Data assembly and parcel-to-block-group aggregation

All variables in the catalog of Section 6 are assembled at their native geographic resolution. American Community Survey variables and CDC PLACES variables that are reported at tract resolution but not at block-group resolution are assigned to all constituent block groups within the tract, that is, the tract value is replicated across each block group nested in the tract. This procedure follows the convention used in the COI 3.0 v3.0 release (Noelke et al., 2024) and acknowledges that the tract-to-block-group disaggregation produces no new information; it simply renders tract-level conditions usable in a block-group-level analytic frame. Parcel-level administrative records (Domain E) are aggregated to block group by spatial join of parcel geometry to the 2020 Census block group boundaries, with the following rules: (1) parcels with centroids falling within a block group polygon are assigned to that block group; (2) count-based variables (code violations, foreclosure filings, vacant parcels, permit issuances) are converted to rates by dividing by the appropriate denominator (housing units, owner units, total parcels, residents) drawn from the ACS or from county parcel counts; (3) the parcel-to-BG aggregation is conducted in ArcGIS with the resulting BG-level rates exported as a tabular file for merge with the survey and modeled-data tables.

Pipeline note (v5): assembly is executed by the thirteen-stage orchestrator (gcnvi_v6_full_pull.py), which adds Stage 10c — dual-source Domain I (recorded incidents plus the provisional AGS cross-validation layer), the Domain J build-out (NTEE nonprofit geocoding, Social Capital Atlas allocation, self-response rates, capped voter-file geocoding), and I7 — with automated verification checks (up to 27) and a run-time manual-downloads registry that documents every gap explicitly.

7.2 Step 2 Margin-of-error filtering

ACS estimates at block-group resolution carry substantially larger margins of error than the same estimates at tract resolution. Spielman et al. (2014) documented systematic patterns of small-area uncertainty in the ACS and recommended that analyses at small-area geographies should screen estimates by their coefficient of variation (CV) or the ratio of the standard error to the point estimate. We propose to flag any variable whose CV exceeds 0.30 at the block-group level as low-reliability and to report it with a flag in the final output. Variables flagged as low-reliability are retained in the catalog and in the analytic procedure, but their inclusion in the final summative composite is conditional on their surviving both the Pearson correlation screen and the regression diagnostic. Flagged variables that survive both screens are included with explicit acknowledgment of their reliability constraint. This protocol follows ASPE (2022) and Spielman et al. (2014).

7.3 Step 3 Z-score standardization

All variables including the four criterion variables and the candidate predictors are converted to z-scores against the Guilford County block-group distribution. Z-scoring proceeds variable-by-variable: for each variable, the mean and standard deviation across all 292 block groups are computed, and the score for each block group is the deviation from the mean expressed in standard deviation units, $z = (x - \mu) / \sigma$. For variables with right-skewed distributions (notably median home value, business density, and several rate variables), we propose to log-transform the variable prior to z-scoring (Tabachnick & Fidell, 2019). The Skewness statistic and the Kolmogorov-Smirnov test of normality are reported for each variable before and after transformation. Variables for which the log transformation does not adequately restore approximate normality are flagged and retained in the analysis with explicit acknowledgment of the residual non-normality.

7.4 Step 4 Pearson correlation screening

Pearson product-moment correlation coefficients are computed between each candidate predictor and each of the four criterion variables. The correlation matrix (approximately 100 predictors by 4 criteria) produces approximately 400 individual correlations. We propose two screening thresholds. The primary threshold is conventional statistical significance at $p < 0.05$ against at least one of the four criteria. Predictors meeting this threshold are advanced to Step 5. The secondary threshold is the more conservative Bonferroni-adjusted threshold of $p < 0.05 / 100 = 0.0005$, accounting for the multiple comparisons across the catalog. We report the variable set under both thresholds, treat the primary threshold as the analytic baseline, and treat the Bonferroni-adjusted set as the sensitivity-analytic robustness check (Field, 2018). Predictors that fail the primary threshold against all four criteria are excluded from further consideration. Their non-correlation is reported as a substantive finding rather than an analytic artifact.

7.5 Step 5 Regression refinement with VIF diagnostics

The set of predictors surviving the Pearson screen is submitted to four parallel ordinary least squares regression models, one for each criterion variable. In each model, the criterion is the dependent variable and the screened predictors are independent variables, with all variables in z-score form so that coefficients are directly comparable in magnitude. Variance inflation factors are computed for each predictor in each model. Following Tabachnick and Fidell (2019), we will adopt $VIF > 5$ as the primary threshold for multicollinearity-driven exclusion, with $VIF > 10$ reported as the more permissive sensitivity-analytic threshold. Predictors exceeding $VIF > 5$ in any of the four models are excluded. Among multicollinear sets, we retain the predictor with the largest absolute zero-order correlation with the criterion in question and exclude the others, following the procedural convention of Hair et al. (2019). The retained predictors are reported with their standardized partial regression coefficients (β), their statistical significance, and their VIF values across all four models.

7.6 Step 6 Composite construction

The final set of retained predictors is summed without further weighting to produce the GCNVI score for each block group. For predictors whose direction in the regression models is negative (higher predictor values associated with lower criterion values), the predictor z-score is multiplied by -1 prior to summation, so that higher final scores uniformly indicate higher predicted vitality. The summative composite carries the methodological commitment that, having survived a correlational and regression-based selection procedure, all retained predictors have demonstrated their contribution to the criterion outcomes and no further differential weighting is required. This is the procedural convention of Mallach (2008, 2012), Sills (2015), and the Stanford Reproducible ADI (Knighton et al., 2016).

7.7 Cronbach's alpha and reliability assessment

Cronbach's α (Cronbach, 1951; Nunnally & Bernstein, 1994) is computed on the final retained predictor set. We will adopt $\alpha \geq 0.70$ as the conventional acceptable threshold for composite reliability, with $\alpha \geq 0.80$ indicating good reliability and $\alpha \geq 0.90$ excellent. We will report the corrected item-total correlations and the "alpha if item deleted" diagnostics for each predictor. Predictors whose deletion would substantially raise α are flagged for reconsideration. Composites with α below 0.70 are flagged as requiring further refinement, possibly by exclusion of items with low item-total correlations or by reconsideration of the predictor set as a whole.

7.8 Visual binning and mapping

The continuous GCNVI z-score sum for each block group is partitioned into five categories using SPSS Visual Binning with cutpoints at Mean $- 1.5$ SD, Mean $- 0.5$ SD, Mean $+ 0.5$ SD, and Mean $+ 1.5$ SD. Assuming an approximately normal distribution of the composite (an assumption verified by inspection of the score histogram and the Shapiro-Wilk test), the resulting categories approximate quintiles. We assign category labels consistent with the Sills (2015) High Point Market Segmentation convention but

inverted in direction so that higher GCNVI values map to higher vitality: Category 1 (Extremely Constrained), Category 2 (Constrained), Category 3 (Functioning), Category 4 (Strong), Category 5 (Highly Vital). The categories are mapped at block-group resolution across all 292 Guilford County block groups, using a sequential red-to-green color ramp (Category 1 red, Category 3 yellow, Category 5 green) following the conventional risk-mapping practice. The continuous z-score sum is retained as a complementary mapping product so that within-category variation remains visible to users who need finer resolution than the five-category schema provides.

7.9 Sensitivity analyses

Three sensitivity analyses accompany the primary GCNVI output. The first repeats Steps 4 through 6 with the COI 3.0 composite excluded from the criterion set, producing a variant index validated against only the three Opportunity Atlas measures; the comparison addresses the partial-circularity concern raised in Section 4.4. The second repeats Steps 4 through 6 with the Bonferroni-adjusted significance threshold ($p < 0.0005$) for the correlation screen; the comparison reports the more conservative variable set. The third repeats Steps 4 through 6 with the $VIF > 10$ multicollinearity threshold; the comparison addresses the more permissive treatment of redundant predictors. Block-group classifications that are stable across all three sensitivity analyses are reported as high-confidence classifications; classifications that change across sensitivity specifications are flagged as analyst-dependent and reported with explicit acknowledgment of the dependence.

7.10 Triangulation reporting

Every block-group GCNVI score is accompanied in the final output by three external composite comparators: the Distressed Communities Index quintile (mapped from ZIP to BG via crosswalk), the Area Deprivation Index national rank (assigned from tract to BG), and the CDC/ATSDR Social Vulnerability Index percentile (assigned from tract to BG). The triangulation reporting operationalizes the first conclusion of the literature review (Sills, 2026): no composite should be treated as a master ranking. Block groups whose GCNVI classification diverges substantively from the external composites are flagged for narrative attention in the final scorecard.

8. Comparing the Method to Alternatives in the Literature

The composite-index literature offers several alternative procedures for constructing neighborhood-level scores, each with distinctive strengths and limitations. We have committed the GCNVI to the outcome-criterion variable selection procedure described in Sections 3 and 7, but the choice deserves explicit justification against the alternatives. The Table below summarizes the comparison; the narrative that follows elaborates on what the table compresses.

Table 2. Comparison of Composite-Index Construction Methods

Method	Strengths	Limitations
Equal-Weight Summative (DCI, SDI; this proposal at Step 6)	Transparent and reproducible. Easily communicated to non-technical audiences. Avoids the opacity of factor-derived weights. Convention in the Distressed Communities Index (EIG, 2025).	Treats all retained predictors as equally important. Does not discount predictors with overlapping content unless multicollinearity is screened separately (as in Step 5 here).
Factor Analysis (PCA, EFA) (ADI v3.2, SVI factors)	Statistically efficient. Identifies latent constructs accounting for shared variance among predictors. Weights are data-driven (Singh, 2003).	Weights opaque to non-technical users. Sample-dependent: weights derived from one population may not apply to another. Petterson (2023) ADI standardization critique and Hannan et al. (2023) reproducibility concerns.
Theory-Derived Weights (County Health Rankings; GCNVI v1.0)	Reflects substantive normative judgment explicitly. Weights survive across vintages and populations. Communication-friendly (UWPHI, 2014; Acevedo-Garcia et al., 2020).	Weights contestable: any weighting represents one analyst's argument about importance. Risk of imposing analyst preferences and reproducing inherited methodological choices.
Outcome-Criterion Selection (Stanford Reproducible ADI; THIS PROPOSAL)	Empirically grounded. Variables retained because they predict outcomes of substantive interest. Updatable as criteria are refreshed. Procedural transparency (Knighton et al., 2016).	Criterion choice embeds a normative position. Variables that do not correlate with criteria but are theoretically important may be excluded. Cannot be used to study the relationship of structural conditions to outcomes (circular for that purpose).
Latent Class / Cluster Segmentation (Mallach Newark/Youngstown; Sills 2015)	Identifies natural typologies rather than enforcing a ranking. Maps directly to differentiated policy intervention (Mallach, 2008, 2018).	Cluster solutions can be unstable across specifications. Cluster meaning requires interpretation. Less suited to county-wide rank comparisons. Visual binning of the present proposal preserves typology output while retaining continuous score for analytic use.
Percentile Ranking (Neighborhood Atlas ADI v3.2)	Familiar and intuitively interpretable. Communicates relative position cleanly.	Information loss in continuous-to-ordinal conversion. Cut-points are arbitrary. The 2016 CHCS Market Segmentation Study (Sills, 2015) used ordinal conversion and lost variance information; this proposal mitigates by retaining continuous z-scores alongside binned categories.

8.1 What the proposed method does best

The outcome-criterion procedure proposed has three principal advantages over the alternatives. First, the variable selection is empirically anchored: variables enter the index because they predict the outcomes the index is built to serve, and variables that do not predict those outcomes are excluded regardless of their theoretical pedigree. This addresses the principal weakness of the theory-derived weighting approach, which is that analysts can disagree about which domains deserve more weight

than others and the disagreement cannot be resolved by reference to the data. Second, the procedure is transparent in a way that factor-analytic procedures are not: every step is documented, every threshold is specified in advance, every sensitivity analysis is reported, and a subsequent analyst can reproduce the result without access to the original analyst's judgment about weighting. Third, the procedure is updatable: as the Opportunity Atlas releases updated estimates, as the COI 3.0 is succeeded by future versions, and as the predictor data are refreshed, the variable selection can be re-run and the variable set can shift in response to changes in the criterion outcomes.

8.2 What the proposed method does less well

Three weaknesses of the procedure deserve explicit acknowledgment. First, the choice of criterion variables is itself a normative position, and the procedure does not escape the choice, it merely relocates it. By choosing the three Opportunity Atlas measures and the COI 3.0 composite as criteria, we have committed the index to a particular substantive position about what makes a neighborhood vital, namely its capacity to generate or sustain economic, social, and developmental opportunity for the children who grow up in it. Other criteria like life expectancy, civic participation, racial integration, environmental quality would produce different variable selections and different rankings. The criterion choice we have made, and the analyst who would use the index for purposes other than the ones the criteria reflect, should be cautious about extending the index beyond its intended use. Second, the procedure cannot be used to test causal claims about the relationship of neighborhood structure to outcomes, because that relationship has been embedded in the selection procedure. The index is a tool for allocation and intervention design, not a research instrument for studying neighborhood effects. Third, the block-group geography introduces small-area uncertainty that the procedural CV-filtering only partially addresses. Some variables will remain in the analysis with margins of error large enough to make individual block-group estimates unreliable, and the user should attend to the reliability flags when interpreting individual block-group scores.

9. Implementation Pathway

9.1 Software and platform

The primary analytic platform is IBM SPSS Statistics version 29, in which all six analytic steps of Section 7 will be executed by the IRI research team under Sills's supervision. The Visual Binning function in SPSS implements the Mean $\pm$ SD cut point procedure natively. Parallel implementation in R could be undertaken as a sensitivity-analytic check, using the *corr* package for the correlation matrix, the *car* package for VIF diagnostics, and the *psych* package for Cronbach's α and McDonald's ω . Block-group geographic processing and the parcel-to-BG aggregation of Domain E variables should be conducted in ArcGIS, with the resulting BG-level tabular data exported for merge into the SPSS analytic file.

9.2 Build sequence and possible timeline

The analytic build proceeds in four phases:

- Phase A Predictor Assembly (June–early July 2026): Compile and load all candidate predictors from federal and partner-verified sources into the SPSS analytic file at block-group resolution. Conduct the margin-of-error filtering of Step 2 and the z-score standardization of Step 3.
- Phase B Criterion-Validated Selection (mid-July 2026): Compute the correlation matrix of Step 4 against the four criterion variables. Run the regression models of Step 5 with VIF diagnostics. Produce the retained predictor set with full procedural documentation.
- Phase C Composite Construction and Diagnostics (late July 2026): Sum the z-scores of retained predictors to produce the GCNVI score for each block group. Compute Cronbach's α . Conduct the three sensitivity analyses of Section 7.9. Generate the visual-binning categorical mapping with assistance from Guilford County GIS Department.
- Phase D Community Validation and Atlas Production (early–mid August 2026): Present block-group classifications to the Coalition of Anchors and/or other community organizations. Document agreements and disagreements between empirical classification and resident judgment. Produce final tract scorecards, the methodology technical appendix, and the data deposit for the Guilford County GIS Department.

9.3 Integration with the Newkirklytics Pilot

Domain K (Newkirknomics economic activity) depends on Newkirklytics field collection for several of its constituent variables and needs further refinement and transparency of sources of these data. The Phase A predictor assembly will populate the federally and publicly available Domain K variables (Census CBP, Nonemployer Statistics, ABS, HMDA, SBA, CDFI, FFIEC CRA) but will leave the Newkirklytics-collected variables (EventureMap-derived business density refinements, NEX Connect borrower-readiness intake, Coalition of Anchors APE self-reporting, incubator/accelerator seat counts, commercial corridor occupancy surveys) as placeholders. The Phase B correlation screening is conducted on the federally and publicly available subset, with the Newkirklytics-collected variables added in a Phase B addendum as their collection matures. This sequencing ensures that the GCNVI production schedule is not gated on the Pilot field-collection schedule while preserving the integration with the Newkirknomics framework as a final-stage refinement. These measures may need to be replaced however if the Newkirklytics data are proprietary and unavailable for future replication.

9.4 Reporting deliverables

The final analytic output comprises five deliverables. The first is the GCNVI Methodology Technical Appendix, which documents every analytic decision, threshold, and sensitivity-analytic result in reproducible detail allowing for repeatable application by the Guilford County GIS team. The second is

the Block-Group Scorecard, produced for each of the 292 Guilford County block groups, reporting the GCNVI score, the visual-binning category, the indicator-level z-scores of the retained predictors, the three external triangulation composites, and a narrative interpretation. The third is the County-Wide Map, displaying the five visual-binning categories across all 292 block groups with the standard red-to-green color ramp. The fourth is the Target-Tract Drilldown, which produces detailed sub-county scorecards for the nine pilot tracts and their constituent block groups. The fifth is the Data Deposit, an SPSS-ready (or ArcGIS or R) data file with all candidate predictors, retained predictors, criterion variables, and composite scores at block-group resolution, later to be deposited in a public-access portal under access controls determined by the Coalition.

10. Limitations and Honest Acknowledgments

Six limitations of the GCNVI deserve direct acknowledgment, in the spirit of the disciplined humility that the composite-index literature has come to require.

10.1 The criterion choice is normative

We have chosen the three Opportunity Atlas measures and the COI 3.0 composite as four criterion variables on the substantive judgment that the purpose of the Atlas is to make legible the structural conditions of neighborhoods that condition the long-run economic outcomes of the children who grow up in them. This is a defensible choice, grounded in Wilson (1987), Sampson (2012), Sharkey (2013), and Chetty and Hendren (2018). But it is a choice, and an analyst working under different substantive commitments could justifiably specify different criteria that would produce different variable selections and different rankings. The GCNVI is not the only defensible composite, and its specific composition reflects its specific purpose.

10.2 Block-group uncertainty constrains some indicators

American Community Survey estimates at the block-group level carry margins of error substantially larger than the same estimates at the tract level (Spielman et al., 2014). The CV-filtering of Step 2 flags variables whose coefficient of variation exceeds 0.30, but flagging is not exclusion: variables that pass the correlation and regression screens with $CV > 0.30$ are retained in the composite with a reliability flag. The user is responsible for attending to those flags when interpreting individual block-group scores, particularly when small differences between block-group rankings are presented.

10.3 Partial circularity with the Child Opportunity Index

The inclusion of COI 3.0 as a criterion variable while also including its component indicators (educational attainment, school proficiency, poverty rate, vacancy rate, and others) as candidate predictors creates partial circularity. A predictor that contributes to COI will, by construction, correlate with COI. The sensitivity analysis described in Section 7.9, wherein the variable selection is repeated

with COI excluded from the criterion set, addresses this concern by reporting which variables survive selection on the three Opportunity Atlas measures alone. Where the sensitivity-analytic specification produces a different variable set from the primary specification, the difference is reported and the user is invited to consider whether the COI dependence is substantively appropriate for the use case at hand.

10.4 The composite cannot test the structure-outcome relationship

Because the variable selection embeds the criterion-predictor correlation in the composite construction, the resulting index cannot be used to test whether neighborhood structural conditions cause the criterion outcomes. The correlation is built into the composite and cannot be tested out of it. This is a substantive constraint on the use of the GCNVI: it is an instrument for allocation and intervention design, not a research tool for the study of neighborhood effects. Analysts who wish to study the causal relationship of neighborhood conditions to child outcomes should use the Opportunity Atlas or COI 3.0 directly, with appropriate causal identification strategies, rather than using the GCNVI as an intermediate construct.

10.5 Parcel-to-block-group aggregation imposes structural assumptions

Domain E variables including code violations, tax delinquency, water and electric utility shut-offs, foreclosure filings, recent sales, and property condition are aggregated from parcel to block group by the spatial join and rate-computation procedures specified in Section 7.1. The procedures assume that the relevant denominator for rate computation (housing units, owner units, total parcels, residents) is itself accurately measured at block-group level, and that the parcel-to-BG spatial join produces an accurate assignment for parcels with centroids near BG boundaries. Both assumptions are defensible, but neither is exact, and the resulting BG-level rates carry an aggregation-introduced uncertainty in addition to the source-data uncertainty.

10.6 The five-category visual binning creates discontinuities

The Mean $\pm$ SD cut point procedure produces five categories that approximate quintiles under the assumption of approximate normality of the composite score distribution. Block groups falling near a cut point may be classified differently from block groups with substantively similar scores on the opposite side of the cut point, creating an artificial discontinuity that the continuous z-score sum does not have. The retention of the continuous score alongside the categorical map (Section 7.8) mitigates but does not eliminate this concern; users who need finer resolution than the five-category schema provides should consult the continuous score.

10.7 Data-source caveats introduced with the v5 expansion

Five caveats attach to the new sources. First, the interim I1–I5 series measures 2014–2018 conditions; recorded crime evolves, and the series is a dated measure of a slowly-moving spatial pattern rather

than of current conditions. Second, all recorded-crime measures capture recorded incidents: reporting propensity itself varies with neighborhood disadvantage and police–community trust, so Domain I is interpreted accordingly. Third, the AGS CrimeRisk layer (I6) is a model that predicts crime partly from socioeconomic covariates already in the candidate set; it is therefore restricted to cross-validation, and the accessible copy is an unofficial re-share of licensed data that must be replaced with a licensed pull before publication. Fourth, the ZIP-level Social Capital Atlas and DCI measures reach block groups through land-area-weighted ZCTA allocation, an approximation that smooths within-ZIP heterogeneity. Fifth, nonprofit density locates organizations at their filing address, which is not their service area (Allard, 2009); the capacity measure (J1e) currently uses BMF-reported revenue rather than 990 program expenses.

10.8 Closing reflection

The procedural rigor of the criterion-validated selection, the multiple sensitivity analyses, the external triangulation comparators, the community-validation step of Phase D, and the explicit acknowledgment of normative choices in the criterion specification all serve to make the argument’s structure visible to the Coalition of Anchors and the Guilford County partners. The resulting index supports differentiated intervention design and resource allocation. It does not supplant the lived knowledge of residents, the substantive judgment of practitioners, or the historical contextualization that the HOLC grades and Race-Income overlay provide. Used in that spirit, as one structured argument among several, the GCNVI will serve the community well.

11. References

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